

17 Feb 2026

$$\phi(t, x_0, \mu) = x_0 + \int_0^t f(\tau, x(\tau); \mu) d\tau$$

$\Downarrow$

$$\frac{\partial \phi}{\partial t} = f(t, x(t), \mu)$$

$$\frac{\partial S}{\partial t} = \frac{\partial}{\partial t} \frac{\partial \phi}{\partial \mu}$$

$$= \frac{d}{d\mu} \frac{d\phi}{dt}$$

$$= \frac{d}{d\mu} f(t, x(t), \mu)$$

$$= \frac{d}{d\mu} f(t, \phi(t, x_0, \mu), \mu)$$

$$= \frac{\partial}{\partial \mu} f(t, \phi(t, \mu), \mu) + \frac{\partial f(t, \phi, \mu)}{\partial \phi} \frac{\partial \phi}{\partial \mu}$$

$S(0, x_0, \mu)$: initial condition = 0

$$\left[\frac{\partial S}{\partial t} \right]_{n \times p} = A(t, \mu) \Big|_{\mu=\mu_0} [S]_{n \times p} + B(t, \mu) \Big|_{\mu=\mu_0}$$

$$[A(t, \mu)]_{n \times n} = \frac{\partial f}{\partial \phi}(t, \phi(t, \mu), \mu)$$

$$[B(t, \mu)]_{n \times p} = \frac{\partial f}{\partial \mu}(t, \phi(t, \mu), \mu)$$

Ex $\dot{x}_1 = -\mu x_2$ $n=2, p=1$

$$\dot{x}_2 = \mu x_1$$

$$f(x, \mu) = \begin{bmatrix} -\mu x_2 \\ \mu x_1 \end{bmatrix}$$

$$A(t, \mu) = \frac{\partial f}{\partial x} = \begin{bmatrix} 0 & -\mu \\ \mu & 0 \end{bmatrix}$$

$$B(t, \mu) = \frac{\partial f}{\partial \mu} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

$$\frac{\partial S}{\partial t} = \begin{bmatrix} 0 & -\mu \\ \mu & 0 \end{bmatrix} S + \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

$$S = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{s}_1 \\ \dot{s}_2 \end{bmatrix} = \begin{bmatrix} 0 & -\mu \\ \mu & 0 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} + \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

$$x(t, \mu_0)$$

$$x(t, \mu) \approx x(t, \mu_0) + [S(t)]_{n \times p} (\mu - \mu_0)_{p \times 1}$$

Ex. $\dot{x}_1 = x_2$

$$\dot{x}_2 = -c \sin x_1 - (a + b \cos x_1) x_2$$

$$a_0 = 1, b_0 = 0, c_0 = 1$$

$$n=2, p=3$$

$$S: \begin{bmatrix} x_3 & x_5 & x_7 \\ x_4 & x_6 & x_8 \end{bmatrix}$$

$$\dot{S} = A(t, \mu)|_{\mu=\mu_0} + B(t, \mu)$$

$$A(t, \mu) = \frac{\partial f}{\partial x}$$

$$= \begin{bmatrix} 0 & 1 \\ -c \cos x_1 + b x_2 \sin x_1 & -(a + b \cos x_1) \end{bmatrix}$$

$$B(t, \mu) = \frac{\partial f}{\partial \mu}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ -x_2 & -x_2 \cos x_1 & -\sin x_1 \end{bmatrix}$$

$$A(t, \mu)|_{\mu=\mu_0} = \begin{bmatrix} 0 & 1 \\ -\cos x_1 & -1 \end{bmatrix}$$

$$B(t, \mu)|_{\mu=\mu_0} = \begin{bmatrix} 0 & 0 & 0 \\ -x_2 & -x_2 \cos x_1 & -\sin x_1 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_3 & \dot{x}_5 & \dot{x}_7 \\ \dot{x}_4 & \dot{x}_6 & \dot{x}_8 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\cos x_1 & -1 \end{bmatrix} \begin{bmatrix} x_3 & x_5 & x_7 \\ x_4 & x_6 & x_8 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ -x_2 & -x_2 \cos x_1 & -\sin x_1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \end{bmatrix} = \begin{bmatrix} x_2 \end{bmatrix} \quad \mu = \begin{bmatrix} a \end{bmatrix} \quad \mu_0 = \begin{bmatrix} a_0 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \\ \dot{x}_7 \\ \dot{x}_8 \end{bmatrix} = \begin{bmatrix} -\sin x_1 - x_2 \\ x_4 \\ -x_3 \cos x_1 - x_4 - x_2 \\ x_6 \\ -x_5 \cos x_1 - x_6 - x_2 \cos x_1 \\ x_8 \\ -x_7 \cos x_1 - x_8 - \sin x_1 \end{bmatrix}$$

$$\begin{bmatrix} b \\ c \end{bmatrix} \quad \begin{bmatrix} b_0 \\ c_0 \end{bmatrix}$$

$$\begin{bmatrix} x_1(t, \mu) \\ x_2(t, \mu) \end{bmatrix} \approx \begin{bmatrix} x_1(t, \mu_0) \\ x_2(t, \mu_0) \end{bmatrix} + \begin{bmatrix} x_3(t, \mu_0) & x_5 & x_7 \\ x_4(t, \mu_0) & x_6 & x_8 \end{bmatrix} \begin{bmatrix} a - a_0 \\ b - b_0 \\ c - c_0 \end{bmatrix}$$

$$\neq x(t, \mu_0 \pm \delta)$$

$$x(t, \mu_0)$$

$$x(t, \mu_0 \pm \delta)$$

μ is scalar

$$\mu \in [\mu_0 - \delta, \mu_0 + \delta]$$

Lyapunov Stability Theory

$$\dot{x} = f(x)$$

$$\text{wlog, } x(0) = 0$$

↓ if eqⁿ is at $x^* \neq 0$

$$\tilde{x} = x - x^* \quad (\Leftrightarrow) \quad x = \tilde{x} + x^*$$

$$\dot{\tilde{x}} = \dot{x} - \dot{x}^* = \dot{x} = f(x) = f(\tilde{x} + x^*)$$

$$\dot{\tilde{x}} = f(\tilde{x} + x^*)$$

$$\dot{x} = -x + 1 = f(x) \quad x^* = 1$$

$$\dot{\tilde{x}} = f(\tilde{x} + x^*)$$

$$= f(\tilde{x} + 1)$$

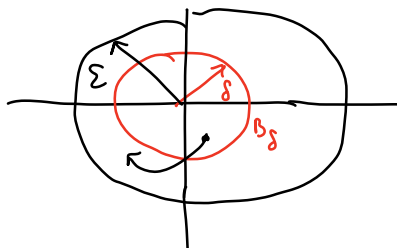
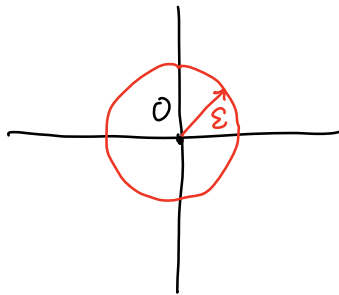
$$= -(\tilde{x} + 1) + 1$$

$$\dot{\tilde{x}} = -\tilde{x} \quad \text{vs.} \quad \dot{\tilde{x}} = -\tilde{x} + 1$$

Lyapunov stability

$x=0$ is stable (in Lyapunov sense) if given $\varepsilon > 0$, $\exists \delta(\varepsilon) > 0$ s.t.

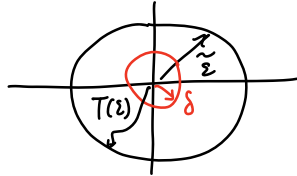
$$|x_0| \leq \delta \Rightarrow |x(t)| \leq \varepsilon \quad \forall t \geq 0$$



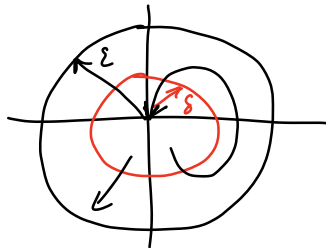
If not stable $\Rightarrow$ unstable

i.e., if $\exists \tilde{\varepsilon} > 0, T(\tilde{\varepsilon}) > 0$ s.t. $\forall \delta > 0$

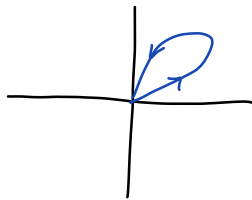
$$|x(t)| > \tilde{\varepsilon} \quad \forall t \geq T$$



$x=0$ is Asymptotically stable (Local) if it is stable and $x(t) \rightarrow 0$ in a neighborhood of $x=0$



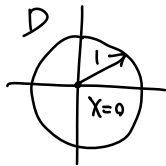
$x=0$ is globally Asymptotically stable (GAS) if stable and $x(t) \rightarrow 0 \quad \forall x(0)$



Lyapunov's Stability Theorem

$$\dot{x} = f(x) \quad x \in \mathbb{R}^n$$

1. Let D be an open, connected subset of $\mathbb{R}^n$ that includes $x=0$.



If there exists a C^1 function

$$V: D \rightarrow \mathbb{R} \quad \text{such that}$$

$$V(0) = 0 \quad \text{and} \quad V(x) > 0 \quad \forall x \in D \setminus \{0\}$$

$$\text{and} \quad \dot{V}(x) := \frac{d}{dt} V(x) = \nabla V(x)^T \frac{dx}{dt}$$

$$= \nabla V(x)^T f(x)$$

$$\leq 0 \quad \forall x \in D$$

then $x=0$ is stable

2. If $\dot{V}(x) < 0 \quad \forall x \in D \setminus \{0\}$

then $x=0$ is asymptotically stable